

# Planets and Kingdoms

Time limit: 1.00 s

Memory limit: 512 MB

A game has  $n$  planets, connected by  $m$  teleporters. Two planets  $a$  and  $b$  belong to the same kingdom exactly when there is a route both from  $a$  to  $b$  and from  $b$  to  $a$ . Your task is to determine for each planet its kingdom.

## Input

The first input line has two integers  $n$  and  $m$ : the number of planets and teleporters. The planets are numbered  $1, 2, \dots, n$ .

After this, there are  $m$  lines describing the teleporters. Each line has two integers  $a$  and  $b$ : you can travel from planet  $a$  to planet  $b$  through a teleporter.

## Output

First print an integer  $k$ : the number of kingdoms. After this, print for each planet a kingdom label between  $1$  and  $k$ . You can print any valid solution.

## Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

## Example

| Input                                         | Output         |
|-----------------------------------------------|----------------|
| 5 6<br>1 2<br>2 3<br>3 1<br>3 4<br>4 5<br>5 4 | 2<br>1 1 1 2 2 |